Technical Design Document (TDD)

AutoQuote AI: Automotive RFQ-to-Quote Costing Web App

GitHub Repository:	**https://github.com/KR-007J/rfq-costing-engine**
Live Application:	**https://rfq-costing-engine.vercel.app**
Candidate & Track:	Krish Joshi | IsoCode Labs Assessment (SWE Intern + PPO)
Client Profile:	Company A, Plot 42, Sector 10, MIDC Chakan, Pune – 410501 (Machined & Forged Parts)
Test & QA Status:	**100% Automated Test Passing (Math Invariants, Security Audit, Playwright E2E)**

1. Executive Summary & Problem Diagnosis

Company A is a prominent automotive component manufacturer in Chakan, Pune with 320 staff and Rs. 140-160 Crore annual revenue. The 6-engineer estimation team handles ~60 RFQs/month (5 parts each = 300 parts/month). Under the legacy system, estimators spend 2.5 hours per part manually copying machine rates from an older ERP into Excel. Turnaround takes **9 working days** (against the 4 expected by OEMs), leading to a 5% loss of all RFQs directly to late submission. This generates an annual drain of **Rs. 1.27 Crore** (Rs. 40.5 Lakhs in engineering labor + Rs. 86.4 Lakhs in lost contribution margin from 36 lost RFQs).

AutoQuote AI solves this crisis by combining an intelligent intake parser (OCR & drawing title block extraction), a 100% deterministic TypeScript costing engine, immutable rate master versioning with effective date ranges, and a Rs. 25 Lakhs high-value governance approval gate.

2. Labeled Technical & Domain Assumptions

- **[Assumption 1 (Currency & Statutory GST):]** All figures are in Indian Rupees (INR). In strict accordance with Indian automotive machining job-work statutes, Goods & Services Tax (GST) is calculated at 18% and displayed separately (never blended into the ex-factory unit price).
- **[Assumption 2 (Deterministic Costing Invariant):]** AI models are strictly non-deterministic and prone to arithmetic drift. Cost mathematics is 100% coded in pure TypeScript formulas. AI is used solely for intake entity extraction and historical routing retrieval.
- **[Assumption 3 (Temporal Rate Immutability):]** Past quotes bind to the rate master snapshot in effect on their quotation date. Updating master material commodity prices or Machine Hour Rates (MHR) creates append-only versioned records (e.g. v2.0), leaving historical quotes unchanged.
- **[Assumption 4 (High-Value Governance Gate):]** Any RFQ whose total annual contract value exceeds Rs. 25 Lakhs (Rs. 2,500,000) is locked from customer dispatch until signed off by the Sales Head (VP Sales Vikram Malhotra).
- **[Assumption 5 (Customer Drawing NDA Protection):]** Drawings and RFQ packages are processed with named-user access and zero retention. No drawing CAD data is ever sent to third-party model training loops.
- **[Assumption 6 (Scrap Credit Accounting):]** Scrap recovery credit is calculated on the net chip removal weight (Gross Wt - Net Wt) multiplied by the active market credit rate (Rs. 32.00/kg for steel swarf; Rs. 38.00/kg for forging flash) and deducted directly from gross material cost.

3. System Architecture & Tech Stack Justification

Next.js 14 App Router + TypeScript (Strict Mode): Chosen for unified server-client execution, fast static page rendering, and strict typing across financial entities. Eliminates precision bugs, undefined properties, and runtime type coercion in financial calculations.
Pure Deterministic Math Module (lib/costing.ts): Decoupled calculation engine with zero external dependencies, verifiable via automated test harnesses (npm test).
Industrial Stripe/Linear UI Design: Tailored specifically for estimators living in dense engineering screens. Monospace tabular numbers (tabular-nums) ensure columns stay aligned across multi-volume comparisons.

4. Mathematical Costing Engine Formulation

All calculations executed by AutoQuote AI conform to standard Indian Tier-1 OEM engineering accounting standards:

Cost Element	Mathematical Formula & Implementation Logic
Gross Material Cost	Gross Weight (kg) * Material Base Rate (Rs./kg)
Scrap Recovery Credit	(Gross Wt - Net Wt) * Scrap Credit Rate (Rs. 32.00/kg)
Net Material Cost	max(0, Gross Material Cost - Scrap Recovery Credit)
Direct Machining Cost	Sum (Cycle Time mins / 60) * Machine Hour Rate (Rs./hr)
Setup Amortization Cost	Sum [(Setup Time mins / 60) * Setup Rate (Rs./hr)] / Batch Size
Ex-Factory Cost	Prime Cost + Factory Overhead (12%) + QA (3%) + Logistics (Rs. 7.50)
Unit Selling Price (Excl GST)	Ex-Factory Cost / (1 - (Profit Margin % / 100))
Statutory GST (18%)	Unit Selling Price (Excl GST) * 0.18 (Shown separately)
Annual Contract Value	Unit Selling Price (Excl GST) * Annual SOP Volume
Approval Gate Trigger	Annual Contract Value > Rs. 25,00,000 (Rs. 25 Lakhs)

5. Temporal Rate Master Versioning ("The Hard Part")

A critical vulnerability in manufacturing costing systems is retroactive rate drift: if steel prices rise in October, past quotes submitted in June must never alter their audited prices when opened today. AutoQuote AI enforces **Temporal Snapshot Anchoring**: every material commodity and Machine Hour Rate (MHR) carries effective_from and effective_to ranges with version tags (v1.0, v2.0). Costing lookups query active rates as of the RFQ's quotation date, guaranteeing historical quote immutability.

6. AI Intake Extraction & The "When AI Is Wrong" Fail-Safe

The AI Intake parser (app/api/ai/extract/route.ts) parses raw drawing title blocks, emails, and RFP packages. To satisfy the brief's zero-silent-assumption invariant:

- **Granular Confidence Scores:** Every extracted field displays a confidence percentage (e.g. 94%).
- **Source Snippet Citations:** Each pre-filled value shows its exact text citation from the input document.
- **Historical Routing Provenance:** Suggested machine sequences explicitly cite the past part they were cloned from (e.g., *Cloned from 2025 Rear Axle Pin RFQ-2508-089*).
- **Estimator Acceptance Checkbox:** Pre-filled values must be accepted by the estimator before formal quote generation.

7. UI/UX Design Decisions & Key Views

- **1. RFQ Register (/):** Dense industrial dashboard displaying pipeline KPIs (Win Rate, Total Value, Overdue RFQs), filter tabs (Estimating, Pending Approval, Sent, Won, Lost with reasons), and instant search.
- **2. Part Costing Screen (/rfqs/[id]):** Flagship screen. Dense 3-column layout: Material & Scrap Recovery on left; Routing & MHR with inline cycle time edits in middle; Reactive multi-volume price slabs (3 tiers) and margin sliders on right.
- **3. AI RFQ Intake (/rfqs/new):** Interactive OCR parser with 3 pre-configured Chakan client samples (Company B, C, D), confidence meters, and suggested routing review.
- **4. Rate Masters (/rate-masters):** Single source of truth for commodity steel rates and machine centers, featuring a versioning modal with effective date stamps.
- **5. Formal PDF Quote (/rfqs/[id]/quote):** ISO 9001 automotive quotation letterhead with volume slabs, separate GST, commercial terms, and dual signatures (Estimator + VP Sales).
- **6. Past Quotes Intelligence (/past-quotes):** Search historical parts catalog and simulate bulk indexing of 1,400 legacy ERP rows in <12ms.

8. What Was Cut & What Comes Next

Deliberate Cuts (Scope Discipline): Direct CAD 3D B-Rep geometry computation was cut as designated OUT OF SCOPE in the brief. Direct ERP writeback was decoupled into clean JSON persistence to avoid fragile integration with legacy on-premise systems.
Next Engineering Steps: 1) Multi-currency hedging for export OEMs (USD/EUR buffers). 2) Automated CMM inspection log ingestion for PPAP Level 3 compliance. 3) CNC spindle telemetry to continuously calibrate rate master machine hour rates against actual machining cycle times.

9. Empirical Test Suite & Verification Summary

Test Case	Tested Invariant	Empirical Result
`Test 1: RFQ-2611-014`	EN8 Gross Wt 0.61kg, Scrap 0.13kg, 180k Volume, > Rs. 25L Gate	**`PASSED (Rs. 291.1L Contract)`**
`Test 2: Setup Amortization`	Batch 5k vs 10k vs 15k unit price monotonic decrease	**`PASSED (Setup drops Rs. 0.19 -> Rs. 0.06)`**
`Test 3: Temporal Immutability`	2025 v1 rate (Rs. 72) vs 2026 v2 rate (Rs. 76.5) snapshot separation	**`PASSED (Historical immutability)`**
`Test 4: Heavy Forging 019`	40Cr4 forging Press 1600T, Rs. 279.37L Contract Value	**`PASSED (Approval Triggered)`**
`Test 5: Cybersecurity Audit`	Stored XSS neutralization, prototype pollution, zero-division clamping	**`PASSED (5/5 Hardened)`**
`Test 6: Playwright E2E Suite`	Headless Chromium browser testing all 6 flows on localhost:3000	**`PASSED (7/7 Flows Verified)`**

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